



Research Paper

Thermal conductivity enhancement of form-stable tetradecanol/expanded perlite composite phase change materials by adding Cu powder and carbon fiber for thermal energy storage

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HIGHLIGHTS

- The paper presents an experimental study on thermal conductivity enhancement.
- Thermal conductivity of composite PCMs is enhanced by adding CuP and CF.
- Method of mixing TCE with PCM is better than implanting TCE into matrix material.
- Composite PCMs have good thermal property for thermal energy storage.

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ABSTRACT

In this study, form-stable tetradecanol (TD)/expanded perlite (EP) composites by adding Cu powder (CuP) and carbon fiber (CF) to enhance thermal conductivities have been prepared via vacuum impregnation method. The thermal conductivity enhancer (TCE) is introduced by two methods of mixing TCE with phase change materials (PCMs) and implanting TCE into matrix materials. The result demonstrates the former is better than the latter in aspects of thermal property, thermal conductivity and controllability, moreover, adding ratio of TCEs gather in more appropriate interval from 2.5% to 3% suitable for the former. For CPCMs, FT-IR results indicate it is no chemical interaction among raw materials but physical combination via two methods. CPCMs prepared by the method of mixing TCE with PCMs have better phase change latent heat and thermal stability by DSC and TGA, while thermal cycling measurements show that form-stable composite PCMs have adequate stability even after being subjected to 200 melting/freezing cycles. Therefore, the method of mixing TCE with PCMs has better thermal property and controllability than that of implanting TCE into matrix materials and the prepared CPCM has great application prospect in solar energy utilization, building material, indoor cooling instrument and so on for thermal energy storage.

1. Introduction

Phase change materials (PCMs) are one type of materials that absorb (release) the thermal energy during the melting (freezing) process. Owing to their excellent energy storage property, PCMs have usually been utilized as one of the most widely implemented materials in the fields of solar energy utilization [1], energy conservation in buildings [2,3], thermal insulation, and thermal regulation [4]. PCMs are divided into different varieties such as paraffin [5,6], fatty acids [7,8],

polyhydric alcohols [9,10], salt hydrate etc. [11,12], and among them many studies have been carried out by researchers in this field.

Thermal energy storage capacity is the most basic property, which depends on the characteristic feature of PCMs. Tetradecanol (TD), belongs to the kind of polyhydric alcohols, is the type of PCM that has high phase change latent (over 200 kJ/kg) and lower phase change temperature (around 32 °C), in addition it possesses of the features of non-corrosive, environmentally friendly and no separation of sub-cooling phase. More importantly, TD is abundantly available and can be

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purchased in low price. Thus, TD has been considered as a preferable PCM that many researchers are willing to studying [13,14].

Meanwhile, thermal conductivity is the most important parameter for the PCM. However, the PCM itself has the low thermal conductivity relatively, which causes the slow heat transfer at the period of absorbing thermal energy and the utilization efficiency of PCM will be enormously reduced, it becomes very necessary to improve the thermal conductivity of experimental PCMs. Currently, there are two methods to resolve the problem, one way is that PCMs are impregnated into the porous materials with high thermal conductivity, another is that the high thermal conductivity materials, such as carbon fiber, Cu powder and so on, are added into the PCMs to achieve the enhancement. The porous materials as a supporting material compositing with the PCMs has been researched widely by plenty of scholars, Yu et al. [15] prepared new Microencapsulated composite PCM by using calcium carbonate as the shell and n-octadecane as PCM to enhance the thermal conductivity. Yuan et al. [16] made use of graphene nanoplatelets and expanded graphite as enhancer to improve the palmitic-stearic acid and gained high thermal conductivity. Mehrali et al. [17] composed the nitrogen-doped graphene and utilized it to load palmitic acid to obtain shape stabilized composite phase change material with high thermal conductivity. In addition, both Zhang [18] and Zhang [19] used expanded graphite as matrix material to enhance the thermal conductivity of PCMs. Despite this, the porous materials with high thermal conductivity have very less variety, and generally have more expensive prices like expanded graphite or grapheme, which has a narrower application and is very difficult to be promoted. Thus, it is particularly important that cheaper and more effective materials are chosen to enhance the conductivity. Due to the reasonable price and high thermal conductivity, Cu powder (CuP) and carbon fiber (CF) usually have been used to implant the PCMs to enhance the thermal conductivity.

In addition, the implantation way of thermal conductivity enhancer (TCE) has important influence on the enhancement effect. Generally, two methods have been used to finish it, the one is by mixing TCE with PCMs, another one is implanting TCE into matrix materials.

In this study, TD is used as thermal energy storage materials, expanded perlite (EP) studied by many researchers as the supporting material, such as Sengul [20], Karaipekli [21] and Sari [22,23], is used to load the composite PCM via vacuum impregnation method, CuP and CF as the TCE are added into composites with two different methods. By the characterization of microstructure, crystal structure, thermal properties and thermal conductivity, it shows that the method of mixing the TCE with PCMs has better thermal property than that of implanting TCE into matrix materials, and also better than the work of Karaipekli [24], Sharma [25], Zhang [28] and so on.

2. Materials and methods

2.1. Materials

The raw materials in the experiment are mainly TD, CF, CuP and EP. Tetradecanol (TD, AR, m.p. 35.8–37.2 °C) was purchased from Tianjin Guangfu Fine Chemical Research Institute. EP and CuP were obtained from Beijing Bo stand Scientific Instruments Co., Ltd. CF was supplied from Shen Zhen Fiber Valley Technology Co. Ltd. The properties of raw materials are presented in detail in Table 1.

Table 1
The basic properties of raw materials including TD, CF, CuP and EP.

Raw materials	Original state	Melting temperature (°C)	Phase change latent heat (J/g)	Thermal conductivity [$\text{Wm}^{-1} \text{K}^{-1}$]
TD	White solid	35.8–37.2	200.2–205.6	0.42–0.46
CF	Black powder	–	–	400–450
CuP	Brown powder	–	–	400 ± 5
EP	Porous mineral	–	–	0.047–0.07

2.2. Sample preparation

The preparation process of CPCMs was divided into two parts by different methods. First, TD-CuP/EP and TD-CF/EP CPCMs preparation were conduct by weighing CuP, CF and TD with ratio from 0% to 4% as showed in Table 2, which was transferred into drying oven and stirred evenly for 12 h at 80 °C to obtain homogeneous mixture. EP as the supporting material was added into the mixture, put into vacuum drying oven with the station of vacuum and taken out with repeated process of absorbing to get the final TD-CuP/EP and TD-CF/EP CPCMs. Fig. 1 shows the preparation process.

Next, TD/EP-CuP and TD/EP-CF CPCMs were prepared via vacuum impregnation method. CuP and CF were added into ethanol respectively. By the ultrasonic dispersion, CuP and CF were dispersed evenly into ethanol in 12 h. EP was added into the mixture, put into vacuum drying oven with the station of vacuum and EP-CuP and EP-CF composites were obtained. Finally, TD was added into EP-CuP and EP-CF taken out with repeated process of absorbing to get the TD/EP-CuP and TD/EP-CF CPCMs. The preparation process was depicted in Fig. 1.

2.3. Sample characterization

The chemical compatibility between the components of the prepared CPCMs was investigated by FT-IR spectroscopy technique. FT-IR spectra were taken on a KBr disk in the frequency range of 4000–400 cm^{-1} by using a FT-IR spectrophotometer (Spectrum 100). Measurements of solid liquid phase change temperatures and latent heat capacities of composite PCMs were carried out by using a DSC instrument (DSC214Polyma) under nitrogen atmosphere and at a heating rate of 10 °C/min. Melting points and freezing points were obtained by drawing a line at the point of maximum slope on the DSC peak. The latent heat was calculated by numerical integration of the peak using the inherent software of the DSC instrument. The thermal stability of the composite PCM was determined by using America TA instruments Q5000. In each case, about 10 mg specimens were heated from 25 °C to 700 °C using a linear heating rate of 10 °C/min under nitrogen atmosphere. The thermal conductivity was test by XFA 500 instrument, according to the standard test method by the Flash Method (ASTM E 1461-01), the specimen was made into the pill that corresponding dimension of is $\phi 12.7 \text{ mm} \times 2 \text{ mm}$.

3. Results and discussion

3.1. FT-IR analyses of prepared samples

Fig. 2 shows the FT-IR spectrum of TD-CuP/EP, TD-CF/EP, TD/EP-CF, TD/EP-CuP composites and the relevant raw materials. As Fig. 2(a) depicted, the peaks at 2918 cm^{-1} and 2850 cm^{-1} represent the symmetrical stretching vibration of $-\text{CH}_3$ and $-\text{CH}_2$ group in pure TD. The deformation-vibration of $-\text{CH}_3$ and $-\text{CH}_2$ group in TD can be recognized at the peak of 1464 cm^{-1} . Besides, the peaks at 1062 cm^{-1} correspond to the in-plane and out-plane bending vibration of the $-\text{OH}$ group of TD. The symmetric peak at 720 corresponds to the swinging vibration of the $-\text{OH}$ functional group. And the peak at 1052 cm^{-1} represents the stretching vibration of $\text{Si}-\text{O}-\text{Si}$ and $\text{Si}-\text{O}-\text{Al}$ in EP. While the peaks of TD-CuP and EP-CuP composites basically are similar

Table 2
Experiment data of preparing the composite PCMs.

Sample	Experiment data (Number setting)									
CuP ratio in TD-CuP composite	0%	0.5%	1%	1.5%	2%	2.5%	3%	3.5%	4%	
	(CuT1)	(CuT2)	(CuT3)	(CuT4)	(CuT5)	(CuT6)	(CuT7)	(CuT8)	(CuT9)	
CF ratio in TD-CF composite	0%	0.5%	1%	1.5%	2%	2.5%	3%	3.5%	4%	
	(CT1)	(CT2)	(CT3)	(CT4)	(CT5)	(CT6)	(CT7)	(CT8)	(CT9)	

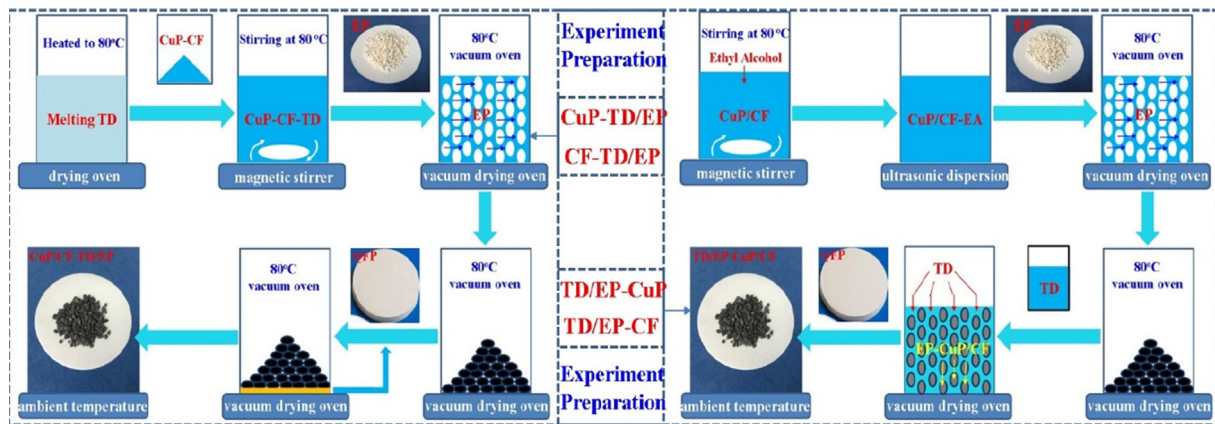


Fig. 1. The preparation process of CuP-TD/EP, CF-TD/EP, TD/EP-CuP and TD/EP-CF CPCMs.

to pure TD and EP because CuP doesn't have any characteristic peak almost, the peaks can't get changed but become relatively gentle. Moreover, the FT-IR spectrum of TD-CuP/EP and TD/EP-CuP composite contains the characteristic peak of CuP, TD and EP respectively. Fig. 2(b) shows the FT-IR spectrum of CF, EP, EP-CF, TD, TD-CF/EP and TD/EP-CF composites, it can also be seen that the TD-CF/EP and TD/EP-CF possess the characteristic peak of CF, TD and EP respectively. Therefore, there is no chemical interaction among the raw materials but physical combination whether the preparation method is mixing CuP and CF with TD or implanting CuP and CF into EP.

3.2. Thermal properties of prepared samples

Fig. 3(a) represent the thermal conductivities of TD-CuP from CuT1 to CuT9 and TD-CF composite from CT1 to CT9. It can be seen that the

thermal conductivity of pure TD is lower than $0.5 \text{ W/(m} \cdot \text{K)}$, the thermal conductivities of both TD-CuP and TD-CF composite trend to improve gradually with the increase of CuP and CF ratio, and the values of right side are bigger than opposite side. Fig. 3(b) showed the EP-CuP and EP-CF, the thermal conductivities of EP-CuP gather from $1.1 \text{ Wm}^{-1} \text{ K}^{-1}$ to $1.3 \text{ Wm}^{-1} \text{ K}^{-1}$, while that of EP-CF are distributed in from $1.3 \text{ Wm}^{-1} \text{ K}^{-1}$ to $1.5 \text{ Wm}^{-1} \text{ K}^{-1}$, which appear irregularity owing to the excessive addition of CuP and CF. By comparison, the method of mixing the CuP and CF with PCMs has better controllability than that of implanting CuP and CF into matrix materials. What's more, for the method of mixing the CuP and CF with PCMs, the thermal conductivity of adding ratio from 0.5% to 3% have improved over 10% than the one of 0%, by comparison of ring ratio whether TD-CuP or TD-CF. But for the adding ratio from 0.5% to 1.5%, the higher growth rates result from small base. Therefore, the adding ratio of from 2.5% to 3%

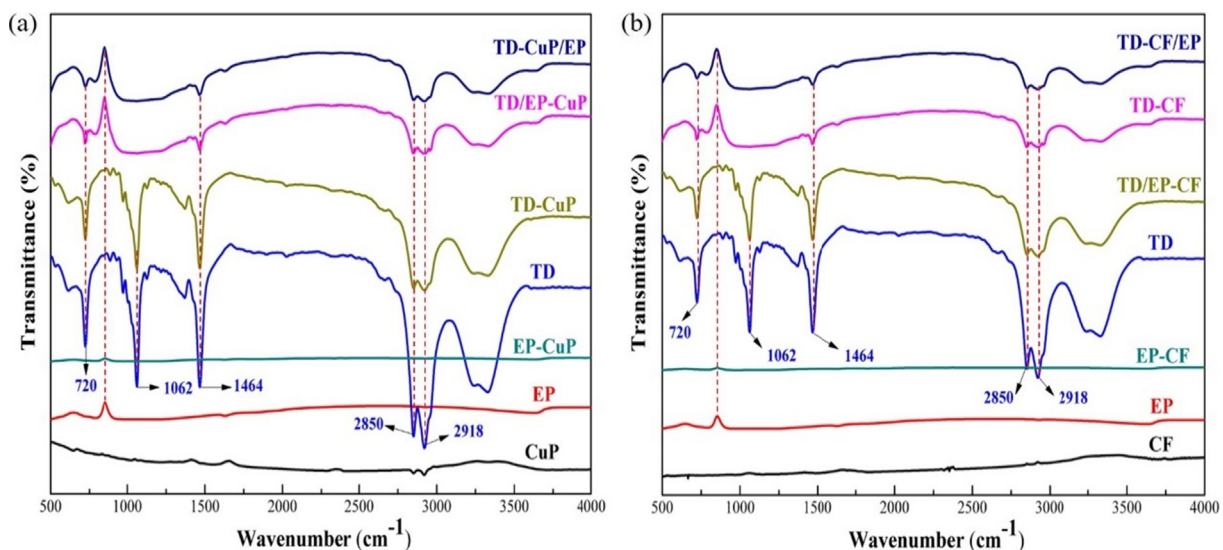


Fig. 2. The FT-IR spectrum: (a) CuP, EP, TD, EP-CuP, TD/EP-CuP, TD-CuP and TD-CuP/EP composites; (b) CF, EP, TD, EP-CF, TD/EP-CF, TD-CF and TD-CF/EP composites.

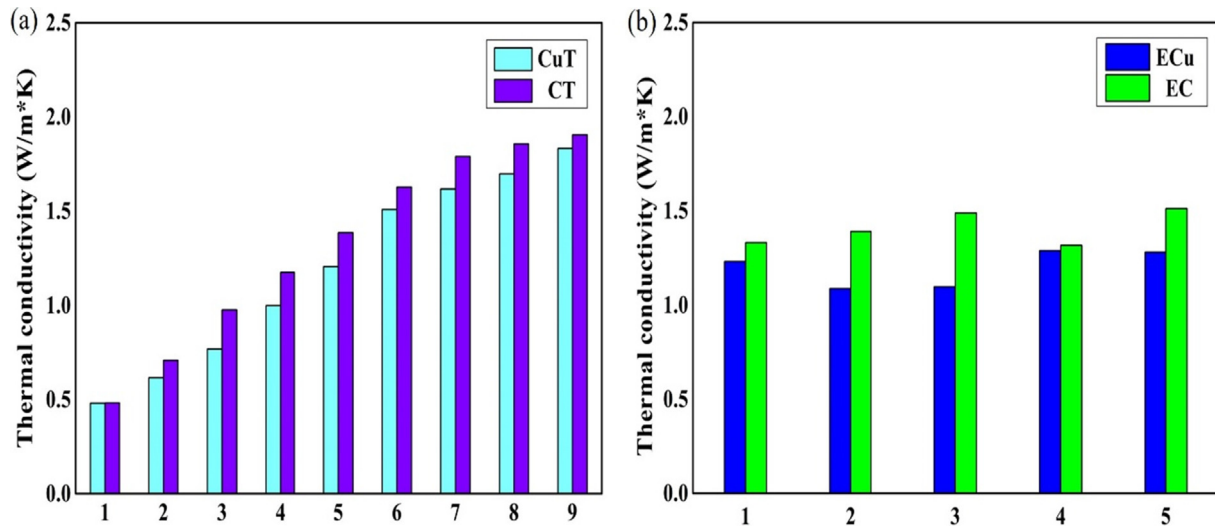


Fig. 3. The thermal conductivities: (a) TD-CuP from CuT1 to CuT9 and TD-CF composite from CT1 to CT9; (b) ECu from ECu1 to ECu5 and ECu from EC1 to EC5.

are ideal.

Fig. 4 represents the melting and freezing processes data of the ratio of TD-CuP/EP, TD-CF/EP, TD/EP-CuP and TD/EP-CF composites. As depicted in Fig. 4(a), the phase change enthalpy of TD/EP is 158.2 kJ/kg and the melting temperature is 36.4 °C. It is noticed the phase change latent heat is apt to drop when the CuP and CF ratio in TD-CuP/EP and TD-CF/EP composite grows gradually, and the melting temperature become lower, the reason is that CuP and CF can accelerate the heat transfer and make TD in composite melt in advance. By the Fig. 4(b), it can be seen that the PCT and PCLH of both TD/EP-CuP and TD/EP-CF composites are changed because the ratio of CuP and CF is uncertain. Thus, the method of mixing the CuP and CF with TD can obtain better thermal property under the control, which is similar to the result of thermal conductivity. Similarly, we can compare the reduction rate of PCLH caused by the introduction of CuP and CF for the method of mixing the CuP and CF with TD, it can be concluded that the adding ratio of 1.5% and 2.5% is for CuP beyond 0.5%, and the adding ratio of 1.5% even rises up to 0.3%, which is attributed to evenly sample when testing. In addition, for CF, the reduction rate by ring comparison below 0.5% are the adding ratio of 1%, 3% and 4%, the adding ratio of 1% happens to rise as well, which caused by making evenly sample. The adding ratio of 4% has lower loss in PCHL, but the PCHL goes down about 4.1% compared with the value of TD/EP. The minimum reduction rate, for CuP is adding ratio of 2.5%, for CF is adding ratio of 3%,

we can see that the difference of 2.5% adding ratio is 1.4 J/g between TD-CuP/EP and TD-CF/EP, while the difference of 3% adding ratio is up to 2.7 J/g when making the same adding ratio into contrast, the smaller difference, the more evenly sample, the more reasonable value. Hence, the adding ratio of 2.5% is ideal.

Fig. 5 shows the DSC curves of pure TD, TD-CuP, TD-CuP/EP composites before and after 200 times thermal cycles. It can be concluded from Fig. 5(a) that the phase change enthalpy is inclined to diminish from TD to TD-CuP/EP composite during melting and freezing process. As displayed in Table 3, TD/EP composite occurs to melt at 36.4 °C with enthalpy value of 158.2 J/g that accounts for 77.7% of pure TD, which can certificate that the load rate of EP is around 77.7%. While the TD-CuP/EP composites melts at 35.5 °C and absorbs 156.7 J thermal energy per gram, also the PCT and PCLH of TD-CF/EP composites are 35.5 °C and 56.7 J/g respectively. Relative to TD/EP composite, TD-CuP/EP composite changes just a little for enthalpy value with CuP added into. To sum up, CPCMs obtain good thermal properties. Fig. 5(b) shows the DSC curves of TD-CuP/EP composites before and after 200 times thermal cycles. It can be seen that both the PCT varies faintly from 35.5 °C to 35.6 °C and the PCLH displays tiny variation about 1.1 J/g after 200 thermal cycles in DSC curve of TD-CuP/EP composites, indicative of a composite with near perfect thermal stability suitable for thermal energy storage applications.

Fig. 6 describes the thermal conductivities of TD, TD-CuP, TD-CF,

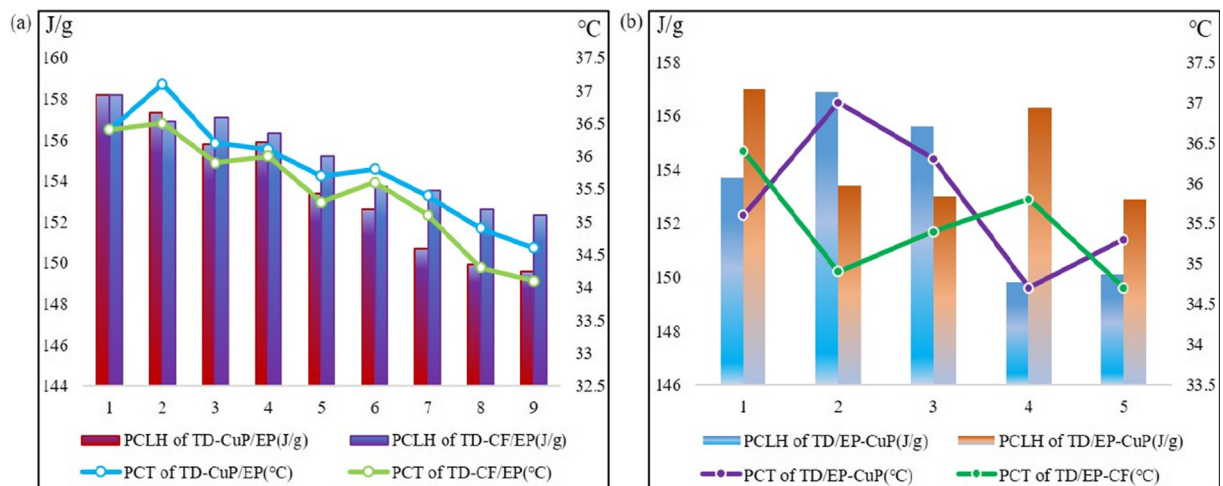


Fig. 4. The thermal properties: (a) TD-CuP from CuT1 to CuT9 and TD-CF composite from CT1 to CT9; (b) ECu from ECu1 to ECu5 and ECu from EC1 to EC5.

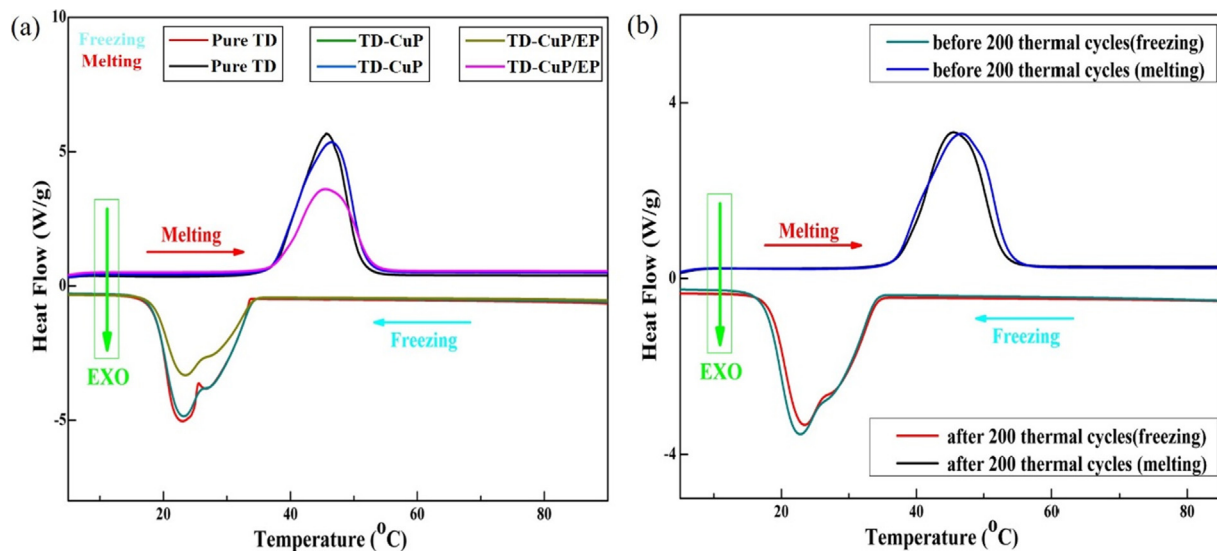


Fig. 5. DSC curves: (a) The melting and freezing processes of pure TD, TD-CuP and TD-CuP/EP composites; (b) The melting and freezing processes of TD-CuP/EP composites and after 200 times thermal cycles.

Table 3

The melting and freezing processes data of TD, TD/EP, TD-CuP, TD-CuP/EP composites before and after 200 thermal cycles.

Sample	Melting process		Freezing process	
	Temperature (°C)	Latent heat (J/g)	Temperature (°C)	Latent heat (J/g)
TD	36.6	203.5	35.8	204.6
TD/EP	36.4	158.2	35.9	158.1
TD-CuP	35.8	199.8	34.9	199.4
TD-CuP/EP	35.5	156.7	36.1	156.1
TD-CuP/EP after 200 thermal cycles	35.6	155.6	35.1	155.8

indicates that EP reduces the thermal conductivity of pure TD. And the thermal conductivity of TD-CuP and TD-CF composite is higher $1.589 \text{ Wm}^{-1} \text{ K}^{-1}$ and $1.617 \text{ Wm}^{-1} \text{ K}^{-1}$, which is enhanced nearly three times by adding CuP and CF compared with pure TD. Besides, the thermal conductivities of TD-CuP/EP and TD-CF/EP composite rise up to $1.463 \text{ Wm}^{-1} \text{ K}^{-1}$ and $1.507 \text{ Wm}^{-1} \text{ K}^{-1}$ in contrast to both TD/EP composite and pure TD, the thermal conductivities are enhanced more twice. It can be concluded that CuP and CF exhibits higher thermal conductivity.

Table 4 presents the thermal properties and conductivities of the TD-CuP/EP and TD-CF/EP CPCMs with comparable PCMs from literature. It can be seen that the prepared TD-CuP/EP and TD-CF/EP CPCMs demonstrate a relatively high thermal enthalpy, a suitable temperature and higher thermal conductivity. Therefore, the prepared CPCMs by mixing the TCE with PCMs demonstrate adequate potential for thermal energy storage systems.

3.3. Thermal stability of prepared samples

Thermogravimetric (TGA) and derivative thermogravimetric (DTG) measurements are pivotal parameters to identify the thermal stability performance of materials. Fig. 7(a) represents the TGA curve of pure TD, TD-CuP, TD-CF, TD-CuP/EP and TD-CF/EP composites, the results of maximum weight loss and temperatures of maximum weight loss are gathered in Table 5. As can be observed, pure TD composite starts to resolve at 135°C and end it at 259.5°C with 100% maximum weight loss, while TD-CuP composite also happens to resolve when heated to 133.3°C and 98% weight losses at the end, the weight losses are due to the decomposition of the PCMs, similarly TD-CuP/EP composite losing 76.3% weight resolves from 134.1°C to 257.3°C . Similarly, the beginning decomposition temperature of TD-CF and TD-CF/EP composites are more than 130°C , and the weight losses reach 97.9% and 76.1%. Compared with pure TD, the weight losses of TD-CuP, TD-CF, TD-CuP/EP and TD-CF/EP composites get smaller and smaller with the increasing of components, which is attributed to that the temperature doesn't reach the decomposing temperature range of CuP, CF and EP. Meanwhile, the 76.1% weight loss of TD-CF/EP composite also proves that the load rate of EP is higher than 76.1% due to the existence of CF and calculation value is about 77.7%, which is very close to the value according to enthalpy calculation. It can be seen from Fig. 7(b) that the thermal degradation temperature trends to reduce gradually with the composites more complex, the thermal degradation of pure TD is at

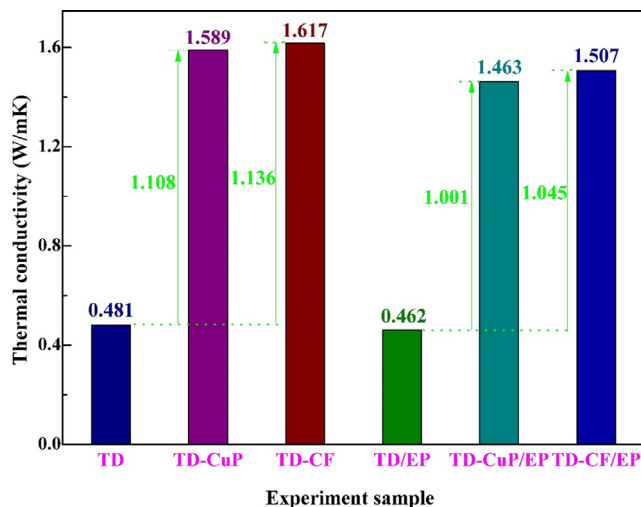


Fig. 6. The thermal conductivities of TD, TD-CuP, TD-CF, TD/EP, TD-CuP/EP and TD-CF/EP composites.

TD/EP, TD-CuP/EP and TD-CF/EP composites, it can be seen clearly that thermal conductivities of pure TD and TD/EP are improved with the CuP and CF added into. As Fig. 6 displayed, the thermal conductivity of pure TD in the solid state is calculated as $0.481 \text{ Wm}^{-1} \text{ K}^{-1}$, while TD/EP composite is calculated as lower $0.462 \text{ Wm}^{-1} \text{ K}^{-1}$, which

Table 4

Comparison of thermal properties and conductivities of prepared CPCM with that of some CPCMs in the literature.

Sample	Melting process		Freezing process		Thermal conductivity (W m ⁻¹ K ⁻¹)	References
	Temperature (°C)	Latent heat (J/g)	Temperature (°C)	Latent heat (J/g)		
Expanded perlite/n-eicosane/Carbon nanotubes (1 wt %)	36.12	157.43	35.5	157.43	0.32	[24]
Palmitic acid-TiO ₂ (5 wt%) composite	61.59	180.03	–	–	0.35	[25]
Capric–palmitic/expanded perlite composite	25.7	88.39	–	–	0.2759	[26]
Lauric–palmitic–stearic acid/expanded perlite composite	31.8	81.5	30.3	81.3	0.44	[27]
Polyethylene glycol-Cu-SiO ₂ composite	53.7	125.2	43.9	110.7	0.513	[28]
TD-CuP/EP composite	35.5	156.7	36.1	156.1	1.463	Present
TD-CF/EP composite	34.6	195.4	35.1	195.8	1.507	Present

upper 223.1 °C, while that of TD-CuP composite is at moderate 222.8 °C and TD-CuP/EP composite at lower 222.6 °C, which are comparatively high in comparison to the indoor temperature. To sum up, it can be concluded that prepared composite PCM has good thermal ability.

Generally, the $T_{\text{heat-resistance-index}}$ (T_{HRI}) is considered as an effective method of determining the heat resistance of materials to ageing under the high temperature. The higher T_{HRI} , the better thermal stability of the materials. The T_{HRI} can be calculated as Eq. (1) by Liang [29] and Gu [30,31].

$$T_{\text{heat-resistance-index}} = 0.49[T_{\text{d1}} + 0.6(T_{\text{d2}} - T_{\text{d1}})] \quad (1)$$

The calculated T_{HRI} is also listed in Table 6. As is shown that after adding CuP and CF, the T_{HRI} of both TD-CuP and TD-CF composites are rising relative to that of pure TD. Similarly does the addition of EP make the T_{HRI} of TD-CuP/EP and TD-CF/EP composites higher compared with TD-CuP and TD-CF composites. We can conclude that the addition of CuP, CF and EP makes the composite PCM gain the better thermal stability.

4. Conclusion

Preparation and characterization of two introduction methods that are mixing TCEs with PCMs and implanting TCE into matrix materials are reported. In CPCMs, TD acts as PCM for thermal energy storage and EP as supporting materials, CuP and CF is used to enhance the thermal conductivity of CPCMs. Taking thermal conductivity and PCHL into consideration, the method of mixing TCEs with PCMs has better controllability than that of implanting CuP and CF into matrix materials both two ways, and the adding ratio of TCEs gather in more appropriate

Table 5

TGA and DTG data of pure TD, TD-CF and TD-CuP/EP composites.

Experiment sample	Temperature of weight loss (°C)			Maximum weight loss (%)
	Beginning	Thermal degradation	Finishing	
TD	135	223.1	259.5	100
TD-CuP	133.3	222.8	258.9	98
TD-CF	132.1	218.9	256.8	97.9
TD-CuP/EP	134.1	222.6	257.3	76.3
TD-CF/EP	133.6	219.5	257.1	76.1

Table 6

Characteristic data of TD, TD-CuP, TD-CF, TD-CuP/EP and TD-CF/EP composites.

Experiment sample	Temperature of weight loss (°C)			$T_{\text{heat-resistance-index}}$
	T_5	T_{30}	T_{50}	
TD	160.53	199.46	212.82	90.1
TD-CuP	169.47	205.33	218.02	93.6
TD-CF	168.96	204.89	217.61	93.4
TD-CuP/EP	169.7	208.76	225.82	94.6
TD-CF/EP	169.2	205.1	219.5	93.5

interval from 2.5% to 3% suitable. For the CPCMs prepared by the method of mixing TCE with PCMs, the crystal structure of TD, CuP, CF and EP remained unchanged in the composite material as verified by FT-IR. Among them, TD-CuP/EP composite melts at 35.5 °C with phase

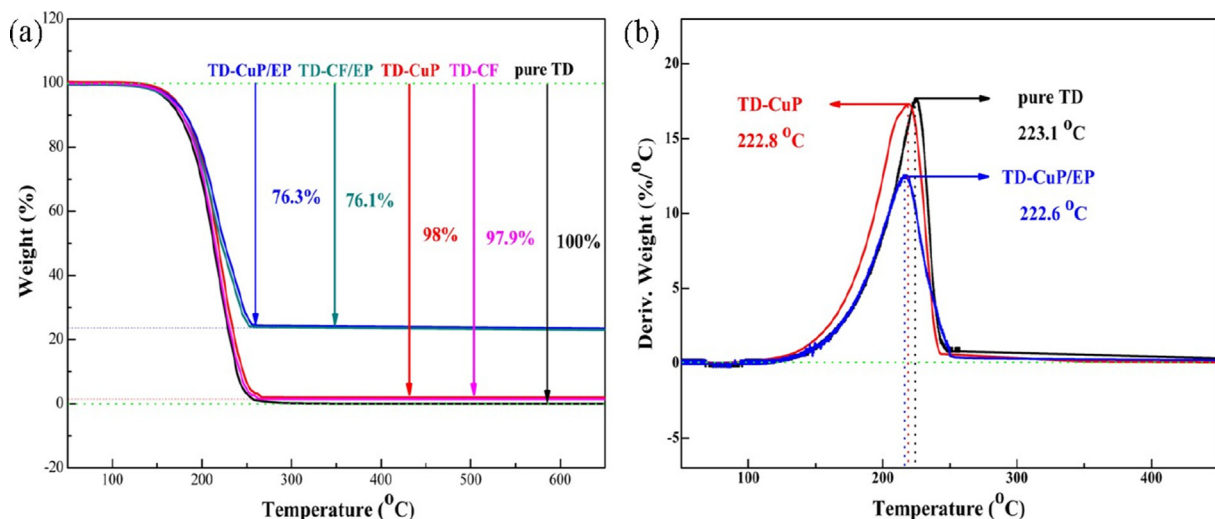


Fig. 7. (a) TGA curves of TD, TD-CuP, TD-CF, TD-CuP/EP and TD-CF/EP composites and (b) DTG curves of TD, TD-CuP and TD-CuP/EP composites.

change enthalpy of 156.7 kJ/kg and freezes at 36.1 °C with phase change enthalpy of 156.1 kJ/kg, while thermal cycling measurements show that the form-stable composite PCM has adequate stability even after being subjected to 200 melting/freezing cycles. What's more, the thermal conductivity is enhanced from $0.481 \text{ Wm}^{-1} \text{ K}^{-1}$ to $1.463 \text{ Wm}^{-1} \text{ K}^{-1}$. Thus, CPCMs by the method of mixing TCE with PCMs have appropriate PCT, good PCHL, high thermal conductivity, excellent thermal stability and better controllability, possess of great application prospect in solar energy utilization, building material, indoor cooling instrument and so on.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.03.140>.

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